Frame Annotation of Legal Reporting Obligations: A Dataset and Baseline

Célestine Lecat[1] Víctor Rodríguez-Doncel[2] Monica Palmirani[1]

[1]Università di Bologna, Bologna, Italy
[2]Universidad Politécnica de Madrid, Madrid, Spain

{celestine.lecat, monica.palmirani}@unibo.it
vrodriguez@fi.upm.es

Abstract

TODO. This paper presents a gold standard dataset of legal provisions annotated with FrameNet frames, focusing on reporting obligations in European Union law. We define a domain-specific frame inventory, describe the annotation methodology, and provide a baseline annotation system built on a large language model. We discuss applications of the resource to legal knowledge representation, compliance analysis, and frame-based reasoning.

1 Introduction

Reporting obligations are among the most pervasive instruments of European Union regulation. Across domains as diverse as banking, insurance, financial markets, data protection, environmental protection and product safety, EU legislation routinely requires regulated actors to transmit specified information to a competent authority, frequently on a recurring basis and within strict time limits. A credit institution must file prudential reports to its supervisor at regular intervals; a data controller must notify the supervisory authority of a personal-data breach without undue delay and, where feasible, within seventy-two hours; an operator must report emissions or incidents to a designated agency. Despite their diversity, such provisions share a recognisable semantic structure: an obligated party, an act of submitting or notifying, a recipient authority, the information to be reported, and temporal parameters expressing deadlines, periodicity or reference points.

The volume and heterogeneity of these obligations has itself become a regulatory concern. Scattered across hundreds of legal acts and drafted in dense natural language, reporting requirements are costly to identify, monitor and keep current, which has motivated efforts both to rationalise them at the policy level and to represent them formally. On the formal side, Palmirani et al. [7] propose an ontology dedicated to reporting requirements in European legislation, modelling the actors involved, the content to be reported and the temporal conditions that govern it. Populating such a model, however, presupposes the ability to locate and interpret the relevant provisions in the legal text itself, a step that remains largely manual.

Frame semantics offers a natural intermediate representation for this step. Prior work on contract law, for example ContractFrames [6], has shown that frame-based structures can bridge natural-language legal text and formal logic by capturing events and their participants in a normalised form. We pursue an analogous strategy for regulatory reporting: we annotate the provisions that impose reporting duties with FrameNet [1] frames, making explicit who must report what, to whom, and when, and we serialise the result as interoperable linked data so that it can feed ontologies such as the one above.

In this paper we make the following contributions. First, we define a domain-specific inventory of FrameNet frames tailored to reporting obligations, adapting general-purpose frames so that their frame elements align with the participants and temporal parameters characteristic of this domain. Second, we present a gold-standard dataset of EU legal provisions annotated with this inventory, together with the annotation methodology. Third, we provide a baseline annotation system built on a large language model and evaluate it against the gold standard. Finally, we discuss applications of the resource to legal knowledge graphs and compliance analysis.

2 State of the Art

FrameNet. FrameNet [1] is a lexical database of English grounded in frame semantics [2], the theory that words evoke structured conceptual frames representing stereotypical situations, events or relations. Each frame defines a set of participants called Frame Elements (FEs), which may be core (semantically obligatory) or non-core (peripheral). FrameNet 1.7 contains over 1,200 frames, 13,000 lexical units and 200,000 annotated example sentences. Frames are interconnected through ontological relations such as *Inheritance*, *Using*, *Subframe* and *Perspective_on*, forming a rich semantic network that supports both lexicographic and computational applications.

Framester. Framester [3] is a large-scale RDF/OWL linked data hub developed at CNR and Università di Bologna that integrates FrameNet with WordNet, VerbNet, BabelNet, DBpedia, Yago and the DOLCE-Zero foundational ontology into a single strongly connected knowledge graph. Frames are formalised as OWL classes, frame elements as properties, and lexical units as linked data resources, enabling full-fledged SPARQL querying and OWL reasoning over frame-based knowledge. Framester exposes a public SPARQL endpoint[1] and its data is available on GitHub.[2] In this work, Framester URIs are used to serialise frame annotations as interoperable Turtle RDF using the NIF standard [4].

2.1 Frame Semantic Parsing

Frame semantic parsing is the task of automatically producing FrameNet-style analyses of text, conventionally decomposed into three interdependent subtasks: frame identification (selecting the frame evoked by a target), argument identification (delimiting the spans that fill its roles) and role classification (labelling those spans with frame elements). A representative recent approach is the double-graph framework of Zheng et al. [8], whose KID model (Knowledge-guided Incremental semantic parser with Double-graph) recasts parsing as incremental graph construction. KID couples two graphs: a *Frame Knowledge Graph*, a static heterogeneous graph that encodes frames and their frame elements together with the ontological relations declared in FrameNet, and a *Frame Semantic Graph*, the per-sentence structure built up incrementally from the text. By letting knowledge-enhanced frame and frame-element representations from the former guide the construction of the latter, the model strengthens the interactions between the three subtasks and the dependencies among arguments that pipeline systems treat in isolation, reporting improvements of up to 1.7 F_1 over the previous state of the art on two FrameNet benchmarks. This separation between an ontological knowledge layer and an instance-level semantic graph is conceptually close to the representation we adopt, although our baseline delegates the parsing itself to a large language model rather than learning a dedicated graph-construction model (Section 4).

2.2 Frame Semantics in Legal Text

The application of frame semantics to legal language has a smaller but growing body of work. Closest to our setting, ContractFrames [6] applies a frame-based representation to contract law, extracting events related to the status of a purchase contract into frames and mapping them to logic clauses of the PROLEG legal-reasoning system through a dedicated Contract Workflow Ontology; it thereby uses frames as an intermediate layer between natural-language contractual text and formal logic, a role analogous to the one we envisage for reporting obligations. More broadly, the enrichment of legal documents with interoperable linguistic annotations has been demonstrated at platform scale by the Lynx project [5], whose service platform processes, enriches and analyses documents from the legal domain and represents the resulting linguistic and semantic annotations as linked data using the NIF standard. Our work follows the same linked-data annotation philosophy, narrowing the focus to FrameNet frames for reporting obligations and serialising them through Framester URIs.

Most directly related to our domain, Palmirani et al. [7] propose a Reporting Requirements Ontology for European legislation, a conceptual model that formalises the obligations imposed on regulated actors to submit information to authorities, capturing the parties involved, the content to be reported, and the temporal conditions such as deadlines and periodicity. Whereas that work operates at the ontological level, defining the classes and properties needed to represent reporting duties, our contribution is complementary and operates at the linguistic level: we annotate the textual provisions themselves with FrameNet frames, yielding the surface evidence from which such ontological structures can be populated. The frame inventory we adopt is in fact designed so that its frames and frame elements align with the participants and temporal parameters of reporting obligations, providing a natural bridge towards ontologies of this kind.

[1] https://w3id.org/framester/sparql
[2] https://github.com/framester/Framester

3 Dataset

TODO.

3.1 Source Corpus

TODO.

3.2 Annotation Methodology

TODO.

3.3 Dataset Statistics

TODO.

4 Baseline Annotation System

TODO.

4.1 System Description

TODO.

4.2 Evaluation

TODO.

5 Applications

TODO.

5.1 Legal Knowledge Graphs

TODO.

5.2 Compliance Analysis

TODO.

6 Conclusion

TODO.

Acknowledgements

TODO.

References

[1] Collin F. Baker, Charles J. Fillmore, and John B. Lowe. The Berkeley FrameNet project. In *Proceedings of the 36th Annual Meeting of the Association for Computational Linguistics and 17th International Conference on Computational Linguistics (COLING-ACL '98)*, pages 86–90, Montreal, Quebec, Canada, 1998. Association for Computational Linguistics.

[2] Charles J. Fillmore. Frame semantics. In *Linguistics in the Morning Calm*, pages 111–137. Hanshin Publishing Company, Seoul, 1982.

[3] Aldo Gangemi, Mehwish Alam, Luigi Asprino, Valentina Presutti, and Diego Reforgiato Recupero. Framester: A wide coverage linguistic linked data hub. In *Proceedings of the 20th International Conference on Knowledge Engineering and Knowledge Management (EKAW 2016)*, volume 10024 of *Lecture Notes in Artificial Intelligence*, pages 243–257. Springer, 2016.

[4] Sebastian Hellmann, Jens Lehmann, Sören Auer, and Martin Brümmer. Integrating NLP using linked data. In *Proceedings of the 12th International Semantic Web Conference (ISWC 2013)*, pages 98–113. Springer, 2013.

[5] Julián Moreno Schneider, Georg Rehm, Elena Montiel-Ponsoda, Víctor Rodríguez-Doncel, Patricia Martín-Chozas, María Navas-Loro, Martin Kaltenböck, Artem Revenko, Sotirios Karampatakis, Christian Sageder, Jorge Gracia, Filippo Maganza, Ilan Kernerman, Dorielle Lonke, Andis Lagzdins, Julia Bosque-Gil, Pieter Verhoeven, Elsa Gomez Diaz, and Pascual Boil Ballesteros. Lynx: A knowledge-based AI service platform for content processing, enrichment and analysis for the legal domain. *Information Systems*, 106:101966, 2022.

[6] María Navas-Loro, Ken Satoh, and Víctor Rodríguez-Doncel. ContractFrames: Bridging the gap between natural language and logics in contract law. In *New Frontiers in Artificial Intelligence: JSAI-isAI 2018 Workshops (JURISIN 2018)*, volume 11717 of *Lecture Notes in Computer Science*, pages 71–84, Cham, 2019. Springer.

[7] Monica Palmirani, Andrea Giovanni Nuzzolese, and Generoso Longo. Reporting requirements ontology for european legislation. In *Electronic Government and the Information Systems Perspective (EGOVIS 2025)*, volume 16049 of *Lecture Notes in Computer Science*, pages 18–32, Cham, 2025. Springer.

[8] Ce Zheng, Xudong Chen, Runxin Xu, and Baobao Chang. A double-graph based framework for frame semantic parsing. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4998–5011, Seattle, United States, 2022. Association for Computational Linguistics.